

Parent Guide — primary / module-07-eleven-multiplication

IKS Curriculum

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Parent Guide — Module 7: The Eleven-Friend Trick

Your child's age: Primary (Grades 3–5) · **Module length:** 10 sessions × 30 min

What your child is learning

A mental-math trick for multiplying any small 2-digit number by 11. The story version:

Every 2-digit number has an “eleven friend.” When they meet, the number grows by adding itself in the middle.

12 meets its eleven friend → **1 (1+2) 2 = 132**.

34 meets its eleven friend → **3 (3+4) 4 = 374**.

The chant: **“Split! Add! Drop!”** 1. **Split** the digits with a gap. 2. **Add** the two digits. 3. **Drop** the sum in the middle.

By end of module your child can do this in under 3 seconds, mentally, for any 2-digit number where the digit-sum is less than 10.

At Day 5 they get a “peek” at what happens when the digit-sum is 10 or more (the “spill-over” — e.g. $73 \times 11 = 803$). They don't have to do this case fully yet — the carry case is Middle-band material.

At Day 6 they meet the Sanskrit name ***Ekādhikena Pūrveṇa*** (lit. “by one more than the previous”) — said once, no quizzing on it. It's the older-kids' name for what they're doing.

What to expect this fortnight

A daily chant: > “Split! the number. Add! the digits. Drop! the sum in the middle. That's the eleven-friend trick!”

Games at home: - The “Eleven Friend” card game (Day 1). - Quick-fire $\times 11$ with a parent or sibling. Call out a 2-digit number; your child shouts back the answer.

A creative project (Days 7–10): Your child will design their own “Eleven Friend” game — cards, board, dice, jumping — and play classmates' games on Day 10.

How to help — and how NOT to help

Helpful: - 5-minute drill before bed. Call out a 2-digit number; your child says the $\times 11$ product. Aim for under 3 seconds per number. - Try the trick yourself. Notice how fast it is once you see the pattern. - For their project: provide materials (cards, paper) and let *them* design the rules.

Not helpful: - Insisting they do “long multiplication” because that’s how you learned. Both methods work; the trick is faster mentally. - Designing their game for them. - Quizzing them on the Sanskrit name. It’s a one-line cultural anchor, not a vocabulary test.

A note on “Vedic mathematics”

You may have heard the term “Vedic mathematics.” The honest framing we use:

- The $\times 11$ trick is a real, useful Indian mental-arithmetic technique.
- It appears in a book called *Vedic Mathematics* by Bharati Krishna Tirthaji, published in 1965 (five years after Tirthaji’s death).
- Whether the trick is genuinely *ancient* Vedic mathematics or a 20th-century synthesis is academically disputed. Scholars (S. G. Dani 1993) note there’s no surviving manuscript supporting the ancient claim.

By Senior band (Grade 9–12) your child will read this debate and form their own view. At Primary band we just teach the technique enthusiastically.

What success looks like

By the end of the module, your child can: - Compute any 2-digit $\times 11$ (no-carry) mentally in under 3 seconds. - Recite “Split, Add, Drop.” - Recognise the Sanskrit name *Ekādhikena Pūrveṇa* when they hear it. - Show classmates a game they designed.

Want to go deeper?

The Middle-band version (Grade 6–8) teaches the full trick including the carry case and the algebraic proof. Preview:
[curriculum/middle/module-07-eleven-multiplication/README.md](#).

10-Day Lesson Plan

Module 7 (Primary) — 10-Day Lesson Plan

At a glance (30-min sessions)

Day	Title	Activity	Output
1	Meet Your Eleven-Friend	Story + first 6 examples	First answers on slate
2	Split, Add, Drop	Chant + practice	Confident chant

Day	Title	Activity	Output
	chant		
3	Friend Game (no-carry)	Card-pair game	First wins
4	Speed Day	Pair quick-fire	Personal best
5	What if the middle is too big?	Hint at carry case	Awe
6	Meet the Sanskrit Name	<i>Ekādhikena Pūrveṇa</i> intro	New word
7	Make-a-Game Day	Design own game	Sketch + first prototype
8	Peek at 3-digit $\times 11$	Teacher demo + watch	Awe
9	Project Day	Build the game	Game ready to play
10	Game Showcase	Play each other's games	Stickers

Daily structure (30 min)

Time	What
4 min	Settle + the “Split, Add, Drop” chant
6 min	Story / new game introduction
15 min	Game / activity
5 min	Sharing / reset

The daily chant

“**Split!** the number. **Add!** the digits. **Drop!** the sum in the middle. That’s the eleven-friend trick!”

Sing it / chorus it with claps. By Day 3 it should be effortless.

Pacing notes

- Games > worksheets at this band. If a worksheet feels slow, drop it and add another game round.
- Day 5 (the carry-case hint) is a “wow” moment — keep it short and mysterious.
- Day 6 names the Sanskrit term ONCE. Don’t drill it.
- Day 8 (peek at 3-digit $\times 11$) is teacher-led demo only. Children watch and gasp; they don’t do it themselves.
- Days 7–10 build to the game showcase. Confirm groups by Day 6.

What children leave with

- Instant recall of any 2-digit $\times 11$ (with digit-sum < 10).
- The “Split, Add, Drop” chant in their bones.
- One Sanskrit word: *Ekādhikena Pūrveṇa* (they don’t need to spell it; they recognise it).
- One game they made and one they played.

- Curiosity about the carry case and 3-digit $\times 11$ for next year.

Homework

Homework — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Hook — What Is the Trick?** · Draw about today's concept. (~5 min)
2. **Day 2 — The Carry Case** · Draw about today's concept. (~5 min)
3. **Day 3 — Why It Works — The Algebra** · Draw about today's concept. (~5 min)
4. **Day 4 — 3-Digit $\times 11$** · Draw about today's concept. (~5 min)
5. **Day 5 — Speed Day** · Draw about today's concept. (~5 min)
6. **Day 6 — Anurūpyena — $\times 12$ to $\times 19$** · Draw about today's concept. (~5 min)
7. **Day 7 — Limits of the Method** · Draw about today's concept. (~5 min)
8. **Day 8 — Tirthaji and Historical Context** · Draw about today's concept. (~5 min)
9. **Day 9 — Mixed Practice** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.